Question 1:

Find the Frobenius power series solution to the differential equation $x^2 \cdot y'' - 2x \cdot y' + 2y = 0$ centered at the singular point $x_0 = 0$.

a) The Frobenius power series solution is ___?___

b) $R =$ Radius of Convergence $= |x_0 - \text{singular point}|$ is ___?___

c) Interval of convergence $= (x_0 - R,\ x_0 + R)$ is ___?___

Question 2:

Find the Frobenius power series solution to the differential equation $2x^2 \cdot y'' - 3x \cdot y' + y = 0$ centered at the singular point $x_0 = 0$.

a) The Frobenius power series solution is __?__

b) $R = \text{Radius of Convergence} = |x_0 - \text{singular point}|$ is __?__

c) Interval of convergence $= (x_0 - R,\ x_0 + R)$ is __?__